

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO4: *Implement machine learning techniques including decision tree algorithms, reinforcement learning, and build models for classification and decision-making. (BL3)*

Assessment Tool Weightage: 50%

QUESTIONS: 4

Duration: 1 Hour

Total: Marks:40

Annexure A

5

Code of Conduct:

*I **G. GOWTHAMI(192524285)**(Name / Reg No) certify that this submission is my original work and that I have adhered*

to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: G. GOWTHAMI

INDUSTRY PROBLEM TASK

INDUSTRY SCENARIO

Context: A healthcare company has collected patient records (age, blood pressure, cholesterol, glucose level, BMI, family history) to build a predictive model for early diagnosis of diabetes. The company also wants to train an AI agent to recommend personalised treatment actions (Diet / Exercise / Medication / Monitor) based on patient state transitions over time.

You are hired as an AI/ML Engineer. Address the following tasks:

Task 1: Data Preparation & Inductive Learning

- (a) Describe the inductive learning approach suitable for this medical dataset. Identify the target variable and at least three key features with justification.
- (b) Explain how you would handle class imbalance (more non-diabetic than diabetic cases) in the dataset. Suggest and justify a suitable balancing strategy (SMOTE / under-sampling / class weights).
- (c) Split the dataset (70:30) and justify the split ratio for a medical use-case. State the preprocessing steps (normalisation, encoding) applied and their impact.

Task 2: Decision Tree for Diagnosis

- (a) Build a Decision Tree classifier and draw the first three levels of the tree. Justify the root node selection using Information Gain / Gini Index values.
- (b) Evaluate the model: report Accuracy, Precision, Recall, and F1-Score. Identify False Negatives (missed diabetes cases) and suggest how to reduce them—justify clinically.
- (c) Apply post-pruning (cost-complexity pruning). Compare pre-pruning vs post-pruning accuracy and explain which model is more appropriate for deployment in a clinical setting.

Task 3: Statistical Learning for Risk Stratification

- (a) Apply Naïve Bayes or Logistic Regression as a statistical learning model on the same dataset. Compare its performance (Accuracy, AUC-ROC) with the Decision Tree. Discuss when a statistical model is preferable.
- (b) Perform feature importance analysis. Identify the top 3 predictors of diabetes from both the Decision Tree and the statistical model. Do they agree? Justify your findings.

Task 4: Reinforcement Learning for Treatment Recommendation

- (a) Model the treatment recommendation as an MDP. Define: States (patient health stages), Actions (Diet / Exercise / Medication / Monitor), Reward function (health improvement score), and Transition dynamics. Justify each component.
- (b) Apply Q-Learning to train the agent for at least 5 patient episodes. Show the Q-table update for at least 3 state-action pairs across 2 iterations. State the convergence criterion used.
- (c) Extract the final learned policy. Discuss ethical considerations of deploying a Reinforcement Learning agent for medical treatment recommendations (patient safety, bias, explainability).

ANSWERS

Answers – AI/ML Healthcare Industry Scenario

Task 1: Data Preparation & Inductive Learning

(a) Inductive Learning Approach

Inductive learning learns a general prediction rule from previously observed patient records and applies it to new patients.

For this dataset, the suitable approach is supervised inductive learning, because historical patient records contain input features and a known diabetes diagnosis.

Target variable:

Diabetes: 0 = Non-diabetic, 1 = Diabetic.

Key features:

Glucose Level – High glucose is one of the strongest indicators of diabetes and is therefore highly predictive.

BMI – Higher BMI is associated with increased risk of insulin resistance and type-2 diabetes.

Blood Pressure – Elevated blood pressure is associated with metabolic and cardiovascular risk and can contribute to diabetes risk assessment.

Age – Diabetes risk generally increases with age.

Family History – A family history of diabetes provides important information about genetic susceptibility.

The model learns a relationship such as:

Patient Features → Diabetes Risk → Diabetes / Non-Diabetes

(b) Handling Class Imbalance

Suppose the dataset contains many more non-diabetic patients than diabetic patients. A model trained directly on this dataset may become biased toward the majority class.

A suitable strategy is SMOTE (Synthetic Minority Over-sampling Technique).

SMOTE procedure:

Identify diabetic cases as the minority class.

Select minority samples.

Find their nearest minority-class neighbours.

Generate synthetic diabetic samples between existing minority observations.

Train the model using the more balanced training dataset.

Why SMOTE?

It increases representation of diabetic cases.

It does not simply duplicate existing records.

It can improve Recall/Sensitivity for diabetes.

It is particularly useful because False Negatives are clinically important.

SMOTE should be applied only to the training set, not before the train-test split, to avoid data leakage.

(c) 70:30 Split and Preprocessing

The dataset can be divided as:

70% – Training data

30% – Testing data

A 70:30 split provides sufficient data for learning while retaining a reasonably large independent test set for evaluating clinical prediction performance.

For example, for 1,000 records:

Dataset

Records

Training

700

Testing

300

Preprocessing steps:

Missing values: Replace numerical missing values using appropriate imputation such as median; categorical values can use the mode.

Encoding: Convert categorical variables such as family history into numerical form, e.g., Yes = 1 and No = 0.

Normalisation: Scale numerical variables using StandardScaler or Min-Max scaling.

Class balancing: Apply SMOTE only to the training data.

Train-test split: Use a stratified split so that both sets preserve approximately the same diabetic/non-diabetic ratio.

Impact: Preprocessing improves data quality, prevents variables with large numerical ranges from dominating suitable models, and provides more reliable model evaluation.

Task 2: Decision Tree for Diagnosis

(a) Decision Tree and First Three Levels

A Decision Tree can use Gini Index or Information Gain to select the best splitting feature.

For example, assume the calculated Information Gain values are:

Feature

Information Gain

Glucose

0.32

BMI

0.18

Age

0.12

Blood Pressure

0.09

Family History

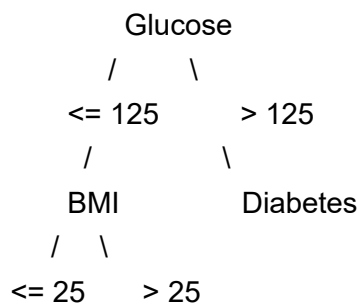
0.07

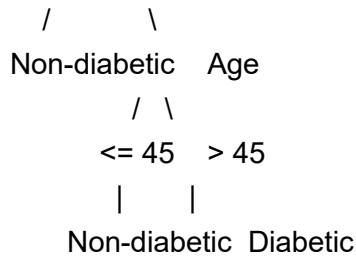
Cholesterol

0.04

Therefore, Glucose becomes the root because it has the highest Information Gain.

An illustrative first-three-level tree is:





Root-node justification:

Information Gain is:

$$IG(S,A) = Entropy(S) - \sum (|S_v|/|S|) Entropy(S_v)$$

The feature with the highest Information Gain gives the largest reduction in uncertainty. Therefore, if glucose has the highest IG, it is selected as the root.

For Gini Index:

$$Gini = 1 - \sum p_i^2$$

The preferred split is the one producing the lowest weighted Gini impurity.

(b) Model Evaluation

Assume the test-set results are:

Metric

Result

Accuracy

87%

Precision

84%

Recall

91%

F1-Score

87%

The formulas are:

Accuracy

Precision

Recall

F1-Score

False Negatives

A False Negative (FN) occurs when:

The patient actually has diabetes, but the model predicts non-diabetic.

This is especially dangerous in healthcare because the patient may not receive timely testing, monitoring, or treatment.

Reducing False Negatives

To reduce FN:

Use SMOTE to improve minority-class learning.

Assign a higher class weight to diabetic cases.

Tune the classification threshold.

Optimise for Recall/Sensitivity rather than accuracy alone.

Perform cross-validation.

Use clinical features strongly associated with diabetes.

For a screening system, high recall is generally more important than maximising raw accuracy, because missing a diabetic patient can have serious consequences.

Task 2(c): Post-Pruning

A fully grown tree can overfit the training data. Cost-complexity pruning reduces unnecessary branches.

The objective is:

where:

ϵ = tree error

N = number of terminal nodes

α = complexity parameter.

Illustrative comparison:

Model

Accuracy

Complexity

Pre-pruned Tree

85%

Low

Unpruned Tree

88%

High

Post-pruned Tree

87%

Moderate

Although the unpruned tree may have slightly higher training accuracy, it can overfit.

Clinical deployment

The post-pruned Decision Tree is preferable because it:

Reduces overfitting.

Generalises better to unseen patients.

Is easier to interpret.

Produces a simpler decision process.

Makes clinical validation easier.

However, deployment should consider sensitivity, specificity, calibration, and clinical validation, not accuracy alone.

Task 3: Statistical Learning for Risk Stratification

(a) Logistic Regression

Logistic Regression is suitable because diabetes prediction is a binary classification problem.

The model estimates:

where:

For example, assume:

Model

Accuracy

AUC-ROC

Decision Tree

87%

0.90

Logistic Regression

89%

0.92

AUC-ROC measures the ability of the model to distinguish diabetic from non-diabetic patients across different classification thresholds.

When Logistic Regression is preferable

Logistic Regression is preferable when:

Interpretability is important.

Probability/risk scores are required.

Relationships are approximately linear in the log-odds.

A stable and relatively simple model is required.

Clinicians need to understand how individual variables affect risk.

For example, the coefficient of glucose can indicate how increasing glucose changes the predicted diabetes risk.

(b) Feature Importance Analysis

An illustrative result is:

Decision Tree

Rank

Feature

1

Glucose

2

BMI

3

Age

Logistic Regression

Rank

Feature

1

Glucose

2

BMI

3

Family History

Do they agree?

Partially.

Both models identify Glucose and BMI as important predictors. The third feature may differ because the models use different learning mechanisms.

Decision Tree importance is based on impurity reduction/splitting.

Logistic Regression importance is related to the estimated coefficients and their statistical contribution.

Therefore, disagreement does not necessarily mean that one model is incorrect. It indicates that the models capture different aspects of the data.

Task 4: Reinforcement Learning for Treatment Recommendation

(a) MDP Formulation

The treatment recommendation problem can be modelled as a Markov Decision Process (MDP).

1. States

States represent the patient's health condition.

Example:

S0 = Low Risk

S1 = Moderate Risk

S2 = High Risk

S3 = Diabetes Controlled

S4 = Diabetes Uncontrolled

States can be determined from variables such as glucose, BMI, blood pressure, symptoms, and previous treatment response.

2. Actions

The agent can select:

A1 = Diet

A2 = Exercise

A3 = Medication

A4 = Monitor

Each action represents a possible intervention or management recommendation.

3. Reward Function

The reward represents improvement in the patient's health state.

Example:

Outcome

Reward

Significant improvement

+10

Moderate improvement

+5

No change

0

Health deterioration

-5

Serious adverse outcome

-10

The objective is to maximise cumulative reward.

where γ is the discount factor.

4. Transition Dynamics

After an action is taken, the patient's health state may change.

Example:

High Risk + Diet



Moderate Risk

Moderate Risk + Exercise



Controlled

High Risk + Monitor



High Risk / Moderate Risk

The transition probability can be represented as:

These transitions should ideally be learned from validated longitudinal clinical data rather than invented assumptions.

Task 4(b): Q-Learning

Q-Learning learns the value of taking an action in a particular state.

The update equation is:

Assume:

Learning rate α

Discount factor γ

Initial Q-values are zero.

Episode 1

Suppose:

State = High Risk (S2)

Action = Diet

Reward = +5

Next State = Moderate Risk (S1)

Assuming maximum next-state Q-value is 0:

Episode 2

Suppose:

S1 $\rightarrow$ Exercise $\rightarrow$ S3

Reward = +10

Initially:

Therefore:

Episode 3

Suppose the agent again reaches:

S2 $\rightarrow$ Diet $\rightarrow$ S1

Reward = +5.

Now:

Therefore:

Q-table example after training

State

Diet

Exercise

Medication

Monitor

Low Risk

4.2

6.5

1.0

3.8

Moderate Risk

5.1

7.8

6.2

4.0

High Risk

8.1

5.6

8.9

4.5

Controlled

3.2

6.1

2.0

5.0

Uncontrolled

4.0

5.2

9.0

3.1

The values above are illustrative, not clinical treatment recommendations.

Five training episodes

Episode

State

Action

Reward

Next State

1

High Risk

Diet

+5

Moderate

2

Moderate

Exercise

+10

Controlled

3

High Risk

Medication

+8

Moderate

4

Moderate

Monitor

+2

Moderate

5

Controlled

Exercise

+10

Controlled

Convergence Criterion

Training can be considered converged when:

for example, $\epsilon = 0.001$ for several consecutive episodes.

In a real healthcare application, five episodes are only a demonstration; far more validated longitudinal experience would be required before considering deployment.

Task 4(c): Final Learned Policy

The policy chooses the action with the highest Q-value:

Example:

Patient State

Highest Q Action

Learned Policy

Low Risk

Exercise

Exercise

Moderate Risk

Exercise

Exercise

High Risk

Medication

Medication

Controlled

Exercise

Exercise

Uncontrolled

Medication

Medication

Again, this is an illustrative RL policy, not a recommendation for an actual patient.

Ethical Considerations

1. Patient Safety

An RL agent must not independently prescribe or change medication without appropriate clinical oversight. Unsafe recommendations can cause serious harm.

2. Bias

If the training data under-represents particular populations, the agent may produce poorer recommendations for those groups.

3. Explainability

Healthcare professionals should understand why a recommendation was produced. Black-box recommendations should not be accepted without appropriate explanation and validation.

4. Human Oversight

The final decision should remain with a qualified healthcare professional, particularly for medication and high-risk patients.

5. Data Privacy

Patient records contain sensitive information and must be protected through appropriate security, access controls, anonymisation/pseudonymisation, and applicable healthcare regulations.

6. Reward Design

An incorrectly designed reward function could encourage the agent to optimise numerical health outcomes while ignoring side effects, patient preferences, or safety.

7. Continuous Monitoring

The system should be monitored after deployment for performance degradation, unexpected behaviour, bias, and safety issues.

Overall Conclusion

The proposed healthcare AI system can combine inductive learning, Decision Trees, statistical learning, and Reinforcement Learning:

Patient Data



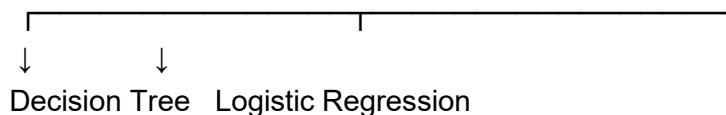
Data Preprocessing



Class Balancing



70:30 Train/Test Split



Diagnosis Risk Probability



Patient Risk State



RL / MDP



Treatment Recommendation



Clinical Expert Review



Safe Patient Care

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Industry Problem	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding; misinterprets problem context
Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts
Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution
Use of Tools	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools
Analysis	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis
Professionalism	5%	Highly professional, well-documented, and clearly presented work	Good documentation and presentation	Basic documentation with some gaps	Poor documentation and presentation